

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Sree BC (92524023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 13-28

18/20

Q. S. A. P

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 10.1.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.1.1.1

However, the workstation is unable to communicate with other systems configured in the network 10.1.1.1/24.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Determining the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IP addresses:

- Device A: 192.168.1.100/24
- Device B: 192.168.1.101/24

Although both devices are connected to the same network, communication fails.

Task

Debug the IP configuration by:

- Identifying the IP addresses and their corresponding subnets.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether both devices belong to the same subnet.
- Identifying the configuration errors.
- Determining the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.

IPv4 Network Debugging

Given IP Address: 192.168.10.65

Subnet Mask: 255.255.255.192 (126)

Default Gateway: 192.168.10.1

(a) Subnet /26 block size = 64

Subnets: 192.168.10.0-63

* 192.168.10.64-127

* 192.168.10.128-191

* 192.168.10.192-255

The IP 192.168.10.65 belongs to 192.168.10.64/26

(b) Network Details

* Network Address: 192.168.10.64

* Broadcast Address: 192.168.10.127

* Valid Host Range: 192.168.

(c) Verify IP and Gateway

* IP address is valid

* Gateway is Invalid (192.168.10.1 belongs to 192.168.10.0/24)

(d) Configuration Error

Default Gateway is in a different Subnet

(e) Optimized Configuration

IP Address: 192.168.10.65

Default Gateway: 192.168.10.66 (or any valid host within 192.168.10.64/26)

(f) Usable Hosts: 64

2) IPv6 Debugging

Given: Device A: 2001::db8::1/64 42.8329/64
Device B: 2001::db8::0:0:1::1/64

(a) Expanded Form

Device A: 2001:0db8:0000:0000:0000:0000:0042:8329

Device B: 2001:0db8:0000:0000:0001:0000:0000:0001

(b) Network Prefix

Both use /64

* Device A Network Prefix: 2001:db8:0:0::/64

* Device B Network Prefix: 2001:db8:0:0::/64

(c) Same Subnet? Yes, both belongs to 2001:db8:0:0::/64

(d) Addressing Error

No prefix mismatch. Communication failure is likely due to incorrect gateway, interface configuration, firewall or Neighbour Discovery Issues.

(e) Optimized Address

* Device A: 2001:db8:0:0::10/64

* Device B: 2001:db8:0:0::20/64

(f) Available Hosts

IPv6/64 provides

$2^{64} = 18,446,744,073,709,551,616$ addresses

11/14 Subnet Optimization (VLSM)

126 * Network 192.168.1.0

* Broadcast: 192.168.1.63

* Host Range: 192.168.1.1 - 192.168.1.62

* Usable Host: ~~62~~ 62

127 * Network: 192.168.1.64

* Broadcast: 192.168.1.95

* Host Range: 192.168.1.65 - 192.168.1.94

* Usable Host: 30

128 * Network: 192.168.1.96

* Broadcast: 192.168.1.111

* Host Range: 192.168.1.97 - 192.168.1.110

~~* Usable Host: 14~~

Unused Address:

* 126 → 64 total → 62 usable

* 127 → 32 total → 30 usable

* 128 → 16 total → 14 usable

Inefficient Allocation

Large department should receive 126.

Small departments should receive 127 or 128

Routing Table Debugging

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP: 192.168.2.100

(a) Longest Prefix Match

Matches: 192.168.2.0/24

(b) Next Hop

10.0.0.3

(c) Destination Subnet

192.168.2.100 belongs to 192.168.2.0/24

(d) Possible Routing Errors

- * Incorrect next-hop address.
- * Missing route on downstream router
- * Interface may be down
- * Routing table may not be updated

2) Optimized Routing

Ensure: # Route points to correct next hop.

Route interfaces are active

Routing entries are verified

3) If No Matching Route Exist

The router uses the Default Route (0.0.0.0/0)

Next Hop: 10.0.0.1

Router